DLNM-LL RCP Simulation

Setup

```r
packages <- c(
  "readxl", "forecast", "dlnm", "splines", "MASS", "astsa"
)
invisible(lapply(packages, library, character.only = TRUE))

find_project_root <- function(start = getwd()) {
  path <- normalizePath(start, mustWork = FALSE)
  for (i in seq_len(8)) {
    if (dir.exists(file.path(path, "Code")) && dir.exists(file.path(path, "Data"))) return(path)
    parent <- dirname(path)
    if (identical(parent, path)) break
    path <- parent
  }
  stop("Set LC_DLNM_BASE_DIR to the project root.")
}
base_dir <- Sys.getenv("LC_DLNM_BASE_DIR")
if (!nzchar(base_dir)) base_dir <- find_project_root()
available_scenarios <- c("rcp26", "rcp85")
default_params <- list(
  scenario = "all",
  scenarios = available_scenarios,
  nsim = 10000,
  simulation_weeks = 1352,
  run_simulation = TRUE
)
rmd_params <- if (exists("params") && !is.null(params)) {
  modifyList(default_params, as.list(params))
} else default_params

if (!is.null(rmd_params$scenario) && !identical(rmd_params$scenario, "all")) {
  scenarios <- rmd_params$scenario
} else {
  scenarios <- rmd_params$scenarios
}
scenarios <- unname(unlist(scenarios))
if (!all(scenarios %in% available_scenarios)) {
  stop(
    "Unknown scenario(s): ",
    paste(setdiff(scenarios, available_scenarios), collapse = ", ")
  )
}

nsim <- rmd_params$nsim
simulation_weeks <- rmd_params$simulation_weeks
```

```r
run_simulation <- isTRUE(rmd_params$run_simulation)

invisible(lapply(list.files(file.path(base_dir, "Code/Function"),
                            pattern = "\\.R$", full.names = TRUE), source))
```

Inputs

```r
historical_inputs <- build_historical_inputs(base_dir)
dat_list <- historical_inputs$dat_list
wave_list <- historical_inputs$wave_list

cb_list <- load_historical_cb_list(
  base_dir,
  lag_days = 21,
  var_df = 3,
  lag_df = 3,
  degree = 3
)
```

Fit DLNM-LL

```r
DLNM_LL_fitting <- DLNM_LL(
  dat_list = dat_list,
  cb_list = cb_list,
  wave_list = wave_list,
  tol = 1e-2,
  max_iter = 20
)
```

```
## converged
```

SARIMA Inputs For Kappa

```r
sarima_inputs <- lapply(scenarios, function(scenario) {
  write_legacy_sarima_inputs(
    model_type = "DLNM_LL",
    scenario = scenario,
    base_dir = base_dir
  )
})
names(sarima_inputs) <- scenarios
sarima_inputs
```

```
## $rcp26
##      model time_series parameter     estimate
## 1  DLNM_LL      Common        ar1  0.757553149
## 2  DLNM_LL      Common         d1  0.000000000
## 3  DLNM_LL      Common        ma1 -0.319981635
## 4  DLNM_LL      Common        ma2  0.009969013
## 5  DLNM_LL      Common       sar1  0.000000000
## 6  DLNM_LL      Common         D1  0.000000000
## 7  DLNM_LL      Common       sma1  0.106674692
## 8  DLNM_LL      Common         S1 52.000000000
```

```
## 9  DLNM_LL      Attiki       ar1  0.403583380
## 10 DLNM_LL      Attiki       ar2  0.254624890
## 11 DLNM_LL      Attiki        d1  0.000000000
## 12 DLNM_LL      Attiki       ma1 -0.044334120
## 13 DLNM_LL      Attiki       ma2 -0.302139040
## 14 DLNM_LL      Attiki      sar1  0.000000000
## 15 DLNM_LL      Attiki        D1  0.000000000
## 16 DLNM_LL      Attiki      sma1  0.143773240
## 17 DLNM_LL      Attiki        S1 52.000000000
## 18 DLNM_LL      Lisbon       ar1  0.336188700
## 19 DLNM_LL        Roma       ar1  1.538669850
## 20 DLNM_LL        Roma       ar2 -0.558647820
## 21 DLNM_LL        Roma        d1  0.000000000
## 22 DLNM_LL        Roma       ma1 -1.388726940
## 23 DLNM_LL        Roma       ma2  0.434116360
## 24 DLNM_LL        Roma      sar1 -0.066841830
## 25 DLNM_LL        Roma        D1  0.000000000
## 26 DLNM_LL        Roma      sma1  0.000000000
## 27 DLNM_LL        Roma        S1 52.000000000
##
## $rcp85
##      model time_series parameter     estimate
## 1  DLNM_LL      Common       ar1  0.757553149
## 2  DLNM_LL      Common        d1  0.000000000
## 3  DLNM_LL      Common       ma1 -0.319981635
## 4  DLNM_LL      Common       ma2  0.009969013
## 5  DLNM_LL      Common      sar1  0.000000000
## 6  DLNM_LL      Common        D1  0.000000000
## 7  DLNM_LL      Common      sma1  0.106674692
## 8  DLNM_LL      Common        S1 52.000000000
## 9  DLNM_LL      Attiki       ar1  0.403583380
## 10 DLNM_LL      Attiki       ar2  0.254624890
## 11 DLNM_LL      Attiki        d1  0.000000000
## 12 DLNM_LL      Attiki       ma1 -0.044334120
## 13 DLNM_LL      Attiki       ma2 -0.302139040
## 14 DLNM_LL      Attiki      sar1  0.000000000
## 15 DLNM_LL      Attiki        D1  0.000000000
## 16 DLNM_LL      Attiki      sma1  0.143773240
## 17 DLNM_LL      Attiki        S1 52.000000000
## 18 DLNM_LL      Lisbon       ar1  0.336188700
## 19 DLNM_LL        Roma       ar1  1.538669850
## 20 DLNM_LL        Roma       ar2 -0.558647820
## 21 DLNM_LL        Roma        d1  0.000000000
## 22 DLNM_LL        Roma       ma1 -1.388726940
## 23 DLNM_LL        Roma       ma2  0.434116360
## 24 DLNM_LL        Roma      sar1 -0.066841830
## 25 DLNM_LL        Roma        D1  0.000000000
## 26 DLNM_LL        Roma      sma1  0.000000000
## 27 DLNM_LL        Roma        S1 52.000000000
```

DLNM Component And Final Simulated Mortality

```r
ll_simulation_outputs <- do.call(rbind, lapply(scenarios, function(scenario) {
  run_ll_rcp_scenario(
    scenario = scenario,
    DLNM_LL_fitting = DLNM_LL_fitting,
    base_dir = base_dir,
    simulation_weeks = simulation_weeks,
    nsim = nsim
  )
}))
```

```
## Running legacy-compatible DLNM-LL RCP simulation for rcp26
```

```
## Running legacy-compatible DLNM-LL RCP simulation for rcp85
```

```r
ll_simulation_outputs
```

```
##   scenario            object
## 1    rcp26 final_LL.sim.rcp26
## 2    rcp85 final_LL.sim.rcp85
##                                                                                                          outpu
## 1 /Users/jiachengabelmin/Desktop/LC_DLNM_paper_repro/Results/Simulated_mortality_rates/final_LL_rcp2
## 2 /Users/jiachengabelmin/Desktop/LC_DLNM_paper_repro/Results/Simulated_mortality_rates/final_LL_rcp8
```